

EDUCATION

Université de Toulouse, Toulouse, France

M.S. Interactions of Computer Science and Mathematics for AI 2025–2026

National Institute of Technology Silchar, Silchar, India

B.Tech. Electronics and Communication Engineering 2021–2025

EXPERIENCE

Institut de Recherche en Informatique de Toulouse (IRIT), France*Research Intern (Full-time)*

May 2024–Mar 2026

May–Jul 2024; Jan–Mar 2025; Sep 2025–Mar 2026

- Researched discourse reasoning and numerical comprehension in large language models under Dr. Nicholas Asher
- Designed and optimized training pipelines for 125M parameter models on multi-GPU HPC clusters, reducing training time through efficient memory management and parallel processing
- Implemented multi-GPU optimization pipeline (up to 12 GPUs) for task-aware layer elimination across 4 LLM families (7B–13B parameters), achieving up to 60% reduction in inference TFLOPs while improving accuracy on 9 benchmarks
- Implemented reproducible experiments using PyTorch, DDP, and automated evaluation pipelines
- Developed a custom fused Triton CUDA kernel replacing softmax attention with Scaled Signed Averaging (SSA), a learnable parameterized score transformation with corrected gradient derivations and log-exp reformulation for numerical stability, achieving 3x speedup over baseline PyTorch attention and matching Megatron-optimized softmax performance across 30K+ training steps on a 114M parameter Nemotron model
- Scaling SSA attention to a 1.3B parameter Nemotron-3 architecture as part of Linagora Labs’ Luciole model series, pretraining from scratch on a 64xH200 GPU cluster (CALMIP KAIROS)

Artificial Intelligence Institute, University of South Carolina*Research Intern (Remote)*

Jan 2023–Dec 2024

- Built automated evaluation pipelines for systematic analysis of 15 large-scale models, processing millions of samples with statistical rigor
- Designed pipelines for statistical testing and task-specific evaluation
- Conducted large-scale experiments on visual paraphrasing and watermarking robustness, highlighting vulnerabilities of current methods
- Published award-winning findings at EMNLP 2023 (Outstanding Paper Award)

PUBLICATIONS

Counter Turing Test CT^2 : AI-Generated Text Detection is Not as Easy as You May Think – Introducing AI Detectability Index*M. Chakraborty, S. M. T. I. Tonmoy, S. M. M. Zaman, K. Sharma, N. R. Barman, C. Gupta, S. Gautam, T. Kumar, V. Jain, A. Chadha, A. P. Sheth, and A. Das*EMNLP 2023 — **Outstanding Paper Award****DIMSUM: Discourse in Mathematical Reasoning as a Supervision Module***N. R. Barman, K. Sharma, N. Asher, and A. Chaturvedi*

SIGDIAL 2025

TELL-TALE: Task Efficient LLMs with Task Aware Layer Elimination*O. Naim, K. Sharma, N. R. Barman, and N. Asher*

(Submitted to ACL 2026)

The Brittleness of Image Watermarking Techniques: Investigating Visual Paraphrasing for De-Watermarking AI-Generated Images
N. R. Barman, K. Sharma, A. Aziz, S. Bajpai, S. Biswas, A. Chadha, V. Sharma, and A. Das
 arXiv 2024

A Transformer-Based Approach to Automate Disease Prediction from Patient Descriptions
N. R. Barman, K. Sharma, and R. Hazra
 IEEE CICT 2023

Addressing Hate Speech: ATLANTIS for Efficient Hate Span Detection
N. R. Barman, K. Sharma, Y. Poddar, A. Vetagiri, and P. Pakray
 FIRE 2023

GRANTS AWARDED	Université de Toulouse ANITI Graduate School Master Excellence Scholarship — Awarded to top 20 out of 500+ candidates 2025–2026
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TECHNICAL SKILLS	Languages: Python, C++, Bash Systems & Performance: CUDA, Triton, Multi-GPU Training, Parallel Programming, Memory Optimization, Profiling (Nsight Systems, NVTX) Algorithms & ML: PyTorch, Transformers, TensorFlow, Scikit-learn, Statistical Analysis Infrastructure: Linux, Docker, Git, Slurm, CI/CD, AWS, GPU Clusters Development: FastAPI, PyTest, Benchmarking
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OPEN SOURCE CONTRIBUTIONS	pytorch/examples: Implemented Vision Transformer from scratch using only native PyTorch libraries, trained and evaluated on CIFAR-10 dataset huggingface/optimum: Removed TF models support from optimum-onnx as it is not supported in Optimum v2
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SELECTED PROJECTS	automanim Nov 2025 <i>FastAPI, Next.js, Manim, LLMs</i> Developed a conversational interface for generating and rendering Manim animations using large language models. Implemented context-aware multi-turn prompting, tree-branching scene history with edit/retry actions, and multi-tab workspaces with auto-save. Integrated pluggable LLM backends with automatic model pulling and live video preview.
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POSITIONS OF RESPONSIBILITY	EMNLP Presenter Dec 2023 <i>Conference Presentation</i> Presented Counter Turing Test (CT²) — Outstanding Paper Award Winner (Resources and Evaluation Track) at EMNLP 2023 Main Poster Session
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POSITIONS OF RESPONSIBILITY	Teaching Assistant/Mentor Jun 2023–May 2025 <i>ML Club, NIT Silchar</i> Conducted Python and Machine Learning sessions for freshmen and sophomores, facilitating peer learning and mentorship
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